

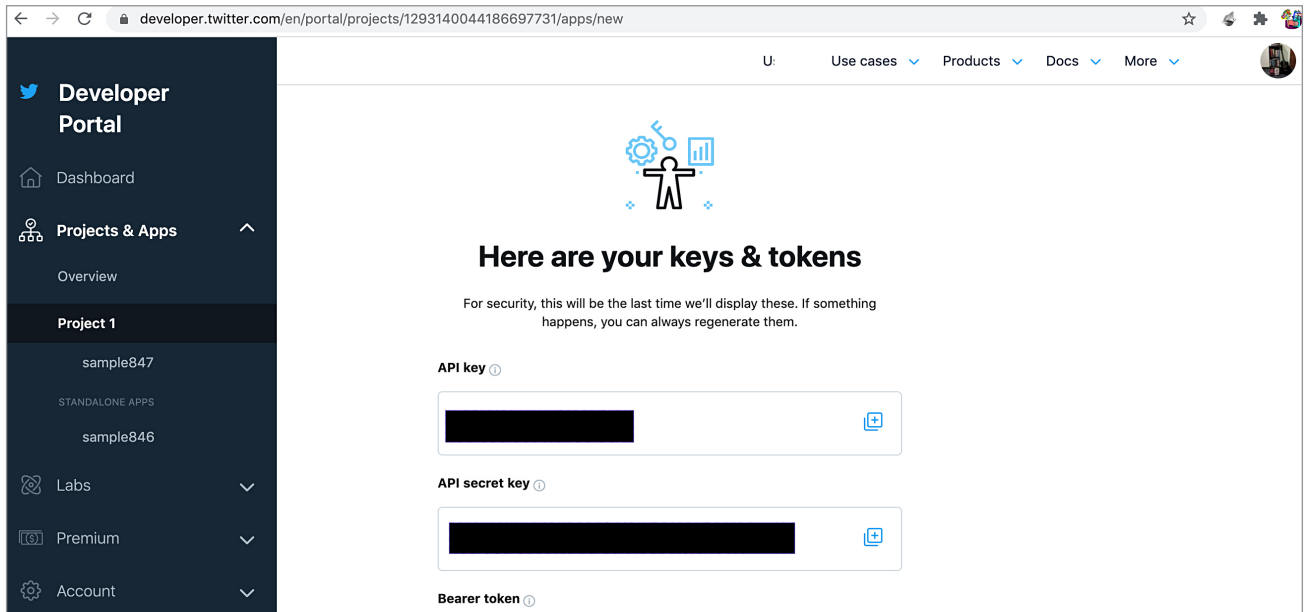


Fig. 2: Adding Access Token and Access Token Secret

Setup steps.

1. *Registering a Twitter App.* In order to use the Twitter API, you first have to register the account as a Twitter application. This will allow you to use the REST interface, which then allows you to interact with Twitter. After registration, go to the *Twitter New App* page and set the name, description, and website to what you want, and then go to *Create Your Twitter Application*.

2. On the page you are taken to next, click on the *Keys and Access Tokens* tab. While here, copy both the consumer key (API Key) and the consumer secret (API Secret); these will be used to connect to the Twitter API. Ignore the other fields as they are not necessary; then click on *Create my access token button*.

3. The page will refresh, after which you should take both the *Access token* and *Access Token Secret* values at the bottom of the page.

4. *Setting and writing the Twitter Bot for Raspberry Pi.*

(i) Make sure Raspberry is up-to-date by running these commands:

```
sudo apt-get update
sudo apt-get upgrade
```

(ii) Install the needed tools to enable the Raspberry Pi Twitter Bot to run. As discussed earlier, we will be using the *Twython* package to interact with the Twitter API. Run the following commands:

```
sudo apt-get install python-setuptools
```

```
sudo easy_install pip
sudo pip install twython
```

(iii) After the installation of the *Twython* package above, create a directory to store your script. Run the following commands:

```
mkdir ~/twitterbot
cd ~/twitterbot
sudo nano TwitterBot.py
```

(iv) Now write the final piece of code and run it:

```
import sys
from twython import Twython

#Define our constant variables, this is all
the data we wrote down in the first part of
the tutorial.
CONSUMER_KEY = '*****YOUR
DATA*****'
CONSUMER_SECRET = '*****YOUR
DATA*****'
ACCESS_KEY = '*****YOUR
DATA*****'
ACCESS_SECRET = '*****YOUR
DATA*****'

#Create a copy of the Twython object with
all our keys and secrets to allow easy
commands.
api = Twython(CONSUMER_KEY, CONSUMER_
SECRET, ACCESS_KEY, ACCESS_SECRET)

#Using our newly created object, utilize
the update_status to send in the text passed
in through CMD
api.update_status(status=sys.argv[1])
```

Replace the values of CON-

SUMER_KEY, CONSUMER_SECRET, ACCESS_KEY, and ACCESS_SECRET above with the values you had got earlier. This code above is to allow communication and writing of text to Twitter. Save and file the code by pressing *Ctrl + X* followed by *Y*, and then pressing *Enter*.

(v) Make the above script executable to enable passing of text to the Python file as an argument. Run:

```
sudo chmod +x TwitterBot.py
```

(vi) You can now test the Twitter Bot by running the following command and checking your Twitter account to see if the tweet was sent:

```
sudo python TwitterBot.py "This is a test
tweet from PiMyLifeUps Twitter bot"
```

In this article, we have taken a look at three useful DIY projects that are also fun and can be done at home with ease. I have shared these projects with the intention of only getting your feet wet with the innumerable applications of the Raspberry Pi. I hope these projects become a good entry point for you to advance to more interesting and fun-filled projects in the field of IoT. **EFY**

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